

Maximum Likelihood : It is the most widely used method of constructing an estimator for unknown parameter θ . In case of large samples, it gives the best estimator of the unknown parameter. Here, maximizing $L(\theta)$ with respect to θ will yield the Maximum likelihood estimator (MLE).

$$L(\theta) = \log L(\theta) = \log \prod_{i=1}^n f(X_i|\theta) = \sum_{i=1}^n \log f(X_i|\theta)$$

Minimum Variance Unbiased Estimator : T is Minimum Variance Unbiased Estimator of $\gamma(\theta)$ if $E_\theta(T) = \gamma(\theta)$ for all $\theta \in \Theta$ and $\text{Var}_\theta(T) \leq \text{Var}_\theta(T')$ for all $\theta \in \Theta$ where T' is any other unbiased estimator of $\gamma(\theta)$

Most Efficient Estimator : If T_1 is the most efficient estimator with variance V_1 and T_2 is any other estimator with variance V_2 , then the efficiency E of T_2 is defined as: $E = V_1 / V_2$. E cannot exceed unity. If $T_1, T_2, \dots, T_n$ are all estimators of $\gamma(\theta)$ and $\text{Var}(T)$ is minimum, then the efficiency of E_i of T_i , ($i = 1, 2, \dots, n$) is defined as: $E_i = \text{Var}(T) / \text{Var}(T_i)$; $E_i \leq 1$, ($i = 1, 2, \dots, n$).

Parameter Space : For a random variable x with p. d. f. $f(x, \theta)$, $\theta \in \Theta$, the set Θ is the set of all possible values of θ and is called the 'parameter space'.

Partial Least Squares : This method is used for estimation when either the number of explanatory (independent) variables are quite large as compared to the number of observations in the data or these explanatory variables are correlated.

Sufficiency : If $T_n = T(x_1, x_2, \dots, x_n)$ is an estimator of a parameter θ , based on a sample $x_1, x_2, \dots, x_n$ of size 'n' from the population with density $f(x, \theta)$ such that the conditional distribution of $x_1, x_2, \dots, x_n$ given T , is independent of θ , then T is sufficient estimator for θ . A sufficient statistic summarises all the information in a sample about the parameter.

30.9 ANSWER OR HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) See Section 30.2
- 2) See Section 30.3

- 3) Here we are given $E(x_i) = \mu$, $v(x_i) = 1 \quad \forall i = 1, 2, \dots, n$.

$$\text{Now } E(x_i^2) = v(x_i) + \{E(x_i)\}^2 = 1 + \mu^2$$

$$E(t) = E\left[\frac{1}{n} \sum_{i=1}^n x_i^2\right] = \frac{1}{n} \sum_{i=1}^n E(x_i^2) = \frac{1}{n} \sum_{i=1}^n (1 + \mu^2) = 1 + \mu^2$$

- 4) We are given $E(X_i) = \mu$, $\text{var}(X_i^2) = \sigma^2$

$$\text{Cov}(X_i, Y_i) = 0 \quad (i \neq j = 1, 2, \dots, n)$$

$$\text{i) } E(t_1) = \frac{1}{5} \sum_{i=1}^5 E(X_i) = (1/5) \cdot 5\mu = \mu \Rightarrow t_1 \text{ is an unbiased estimator of } \mu.$$

$$\text{ii) } E(t_3) = \mu \Rightarrow \lambda\mu = 0 \Rightarrow \lambda = 0$$

$$v(t_1) = [v(X_1) + v(X_2) + v(X_3) + v(X_4) + v(X_5)] / 25 = \sigma^2 / 5$$

$$v(t_2) = [v(X_1) + v(X_2)] / 4 + v(X_3) = 3\sigma^2 / 2$$

$$v(t_3) = [4v(X_1) + v(X_2)] / 9 = 5\sigma^2 / 9 \quad (\text{since } \lambda = 0)$$

Since $v(t_1)$ is the least, t_1 is the best estimator (in the sense of least variance) of μ .

$$5) \quad L(x, \theta) = \prod_{i=1}^n f(x_i, \theta) = \theta^n \prod_{i=1}^n (x_i^{\theta-1})$$

$$= \theta^n \left(\prod_{i=1}^n x_i \right)^{\theta} \cdot \frac{1}{\left(\prod_{i=1}^n x_i \right)} = g(t_1, \theta) \cdot h(x_1, x_2, \dots, x_n), \text{ (say).}$$

Hence by Factorization theorem, $t_1 = \prod_{i=1}^n x_i$, is sufficient estimator for θ .

Check Your Progress 2

- 1) See Section 30.5
- 2) See sub-section 30.5.3
- 3) Let p be the probability of getting heads. Then the probability of getting HHT is $p^2(1-p)$. Taking the derivative and setting equal to zero gives $p = (2/3)$

$$4) L(x_1, \dots, x_n | \theta) = \prod_{i=1}^n \frac{\theta x_i^{\theta-1}}{2^\theta}$$

$$\ln L(x_1, \dots, x_n | \theta) = \sum_{i=1}^n (\ln \theta + (\theta - 1) \ln x_i - \theta \ln 2)$$

$$\frac{\partial}{\partial \theta} \ln L(x_1, \dots, x_n | \theta) = \sum_{i=1}^n \left(\frac{1}{\theta} + \ln x_i - \ln 2 \right) = 0$$

$$\frac{n}{\hat{\theta}} + \sum_{i=1}^n \ln x_i - n \ln 2 = 0$$

$$\frac{n}{\hat{\theta}} = n \ln 2 - \sum_{i=1}^n \ln x_i$$

$$\hat{\theta}_{MLE} = \frac{n}{n \ln 2 - \sum_{i=1}^n \ln x_i}$$

30.10 EXERCISES

Q1. Let $X_1, X_2, \dots, X_n$ be a random sample from a Cauchy population:

$$f(x, \theta) = \frac{1}{\pi} \cdot \frac{1}{1 + (x - \theta)^2}; \quad -\infty < x < \infty, \quad -\infty < \theta < \infty.$$

Examine if there exist sufficient estimate for θ .

Ans. $L(x, \theta) = \prod_{i=1}^n f(x_i, \theta) = \frac{1}{\pi^n} \cdot \prod_{i=1}^n \left[\frac{1}{1 + (x_i - \theta)^2} \right] \neq g(t_1, \theta) \cdot h(x_1, x_2, \dots, x_n).$

Hence by Factorization Theorem, there is no single statistic, which alone is sufficient for θ .

However, $L(x, \theta) = k_1(X_1, X_2, \dots, X_n, \theta) \cdot k_2(X_1, X_2, \dots, X_n) \Rightarrow$ The whole set $(X_1, X_2, \dots, X_n)$ is jointly sufficient for θ .

Q2. Given that baby weights are normally distributed with mean μ and standard deviation σ , with the three babies weighing 8, 9, 10 ounces, what is the maximum likelihood for μ ?

Ans. The p.d.f for a normal distribution is $\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$. So, we want to maximize

$$\frac{1}{\sigma^3 \sqrt{2\pi}^3} e^{-(8-\mu)^2/2\sigma^2 - (9-\mu)^2/2\sigma^2 - (10-\mu)^2/2\sigma^2}$$

For μ and take the logarithmic derivative and set that equal to 0.

Q3. The number of sixes made during a match is Poisson distributed. Last week, the number of sixes made was 14, 27, 25, and 12. Calculate the maximum likelihood estimate for the parameter λ .

Ans. The likelihood function is $L(\lambda | 14, 27, 25, 12) = P(14, 27, 25, 12 | \lambda)$
 $= \frac{\lambda^{14} e^{-\lambda}}{14!} \cdot \frac{\lambda^{27} e^{-\lambda}}{27!} \cdot \frac{\lambda^{25} e^{-\lambda}}{25!} \cdot \frac{\lambda^{12} e^{-\lambda}}{12!} = \frac{\lambda^{78} e^{-4\lambda}}{14! 27! 25! 12!}$. Take the derivative and set it equal to 0

Q4. If X_i are normally distributed random variables with mean μ , then:

$$\hat{\mu} = \frac{\sum X_i}{n} = \bar{X}$$

is the maximum likelihood estimators of μ . Is the MLE unbiased for the parameter?

Ans. If X_i is a normally distributed random variable with mean μ and variance σ^2 , then $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$. Therefore:

$$E(\bar{X}) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \frac{1}{n} \sum_{i=1}^n \mu = \frac{1}{n} (n\mu) = \mu$$

Therefore, the MLE of μ is unbiased.

UNIT 31 HYPOTHESIS TESTING

Structure

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31.0 OBJECTIVES

After going through this unit, you should be able to

- form the null hypothesis and the alternative hypothesis;
- determine the critical region and the level of significance;
- specify the confidence interval and confidence limits;
- calculate the critical values in one-tailed and two-tailed tests;
- distinguish between Type I and Type II errors;
- identify the optimum test under different situations;
- compare two univariate normal distributions; and
- solve the problems relating to hypotheses testing.

31.1 INTRODUCTION

After understanding the theory of estimation and various methods of estimation in the previous unit, now we are prepared to learn further about statistical inference. We generally have some information or hypothetical values of the parameters and it is required to test how far the hypothesis is tenable in the light of the information provided by the sample. The procedure which enable us to decide whether a certain hypothesis is true or not is called the test of hypothesis or test of significance. In this unit, we will learn about the basics like null hypothesis, alternative hypothesis, how to determine the critical region and the level of significance, how to specify the confidence interval and confidence limits, etc. You will be able to understand the difference between Type I and Type II errors and also know about the optimum test under different situations. Further there will be a discussion about the comparison between two univariate normal distributions and how to recognize the problems relating to a bivariate normal distribution.

31.2 TESTS OF SIGNIFICANCE

A very important aspect of the sampling theory is the study of tests of significance, which enables us to decide on the basis of the sample results, if (i) the deviation between the observed sample statistic and the hypothetical parameter value, or (ii) the deviation between two independent sample statistics, is significant or might be attributed to chance or the fluctuations of sampling. Hypothesis testing is always about a population parameter, not about a sample statistic.

Since, for large 'n', almost all the distributions, e.g., binomial, poisson, negative binomial, hypergeometric, t, F, chi-square, can be approximated very closely by a normal probability curve, we use the 'normal test of significance' for large samples. Some of the well-known tests of significance for studying such differences for small samples are t-test, F-test and Fisher's z-transformation.

31.2.1 Null Hypothesis

The technique of randomization used for the selection of sample units makes the test of significance valid. For applying the test of significance, we first set up a hypothesis – a definite hypothesis of no difference. It is called as the 'null hypothesis' and usually denoted by H_0 . According to Professor R. A. Fischer, null hypothesis is the hypothesis, which is tested for possible rejection under the assumption that it is true.

For example, in case of single statistic, H_0 will be that the sample statistic does not differ significantly from the hypothetical parameter value and in the case of a comparison between two statistics, H_0 will be that the sample statistics do not differ significantly. Having set up the null hypothesis, we compute the probability P that the deviation between the observed sample statistic and the

hypothetical parameter value might have occurred due to fluctuations of sampling. If the deviation comes out to be significant (as measured by a test of significance), null hypothesis is refuted or rejected at the particular level of significance adopted and if the deviation is not significant, null hypothesis may be retained at that level.

31.2.2 Alternative Hypothesis

Any hypothesis, which is complementary to the null hypothesis is called an alternative hypothesis, usually denoted by H_1 .

For example, if we want to test the null hypothesis under the three scenarios related to the population mean μ , specified by (say $\mu = \mu_0, \mu \leq \mu_0, \mu \geq \mu_0$) then the alternative hypothesis could be

- i) $H_0 : \mu = \mu_0, H_1 : \mu \neq \mu_0$
- ii) $H_0 : \mu \leq \mu_0, H_1 : \mu > \mu_0$
- iii) $H_0 : \mu \geq \mu_0, H_1 : \mu < \mu_0$

The alternative hypothesis in (i) is known as ‘two-tailed alternative’ and the alternatives in (ii) and (iii) are known as ‘right-tailed alternative’ and ‘left-tailed alternative’, respectively. The setting of alternative hypothesis is very important since it enables us to decide whether we have to use a single-tailed (right or left) or two-tailed test.

31.2.3 Critical Region and Level of Significance

A region (corresponding to a statistic t) in the sample space S that amounts to rejection of H_0 is termed as ‘critical region’. If ω is the critical region and if $t = t(X_1, X_2, \dots, X_n)$ is the value of the statistic based on a random sample of size ‘ n ’, then $P[t \in \omega \mid H_0] = \alpha$ and $P[t \in \omega' \mid H_1] = \beta$, where ω' is the complementary set of ω , called the ‘acceptance region’. We have $\omega \cup \omega' = S$ and $\omega \cap \omega' = \phi$. The probability of α that a random value of the statistic t belongs to the critical region is known as the ‘level of significance’. In other words, level of significance is the size of the type I error (or the maximum producer’s risk). The levels of significance usually employed in testing of hypothesis are 5% and 1%. The level of significance is always fixed in advance before collecting the sample information.

31.2.4 Confidence Interval and Confidence Limits

Let x_i , ($i = 1, 2, \dots, n$) be a random sample of ‘ n ’ observations from a population involving a single unknown parameter θ (say). Let $f(x, \theta)$ be the probability function of the parent distribution from which the sample is drawn and let us suppose that this distribution is continuous. Let $t = t(x_1, x_2, \dots, x_n)$, a function of the sample values, be an estimate of the population parameter θ , with the sampling distribution given by $g(t, \theta)$.

Having obtained the value of the statistic t from a given sample, the problem is, “Can we make some reasonable probability statements about the unknown parameter θ in the population, from which the sample has been drawn?” This question is very well answered by the technique of ‘confidence interval’ due to Neyman and is obtained below.

We choose once for all some small value of α (5% or 1%) and then determine two constants, say, c_1 and c_2 such that $P [c_1 < \theta < c_2] = 1 - \alpha$. The quantities c_1 and c_2 , so determined, are known as the ‘confidence limits’ and the interval $[c_1, c_2]$ within which the unknown value of the population parameter is expected to lie, is called the ‘confidence interval’ and $(1 - \alpha)$ is called the ‘confidence coefficient’. Thus, if we take $\alpha = 0.05$ (or 0.01), we shall get 95% (or 99%) confidence limits.

How to find c_1 and c_2 ?

Let T_1 and T_2 be two statistics such that $P (T_1 > \theta) = \alpha_1$ and $P (T_2 > \theta) = \alpha_2$ where α_1 and α_2 are constants independent of θ . So, it can be written that $P (T_1 < \theta < T_2) = 1 - \alpha$ where $\alpha = \alpha_1 + \alpha_2$. Statistics T_1 and T_2 may be taken as c_1 and c_2 as defined in the last section.

For example, if we take a large sample from a normal population with mean μ and standard deviation σ , then $Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$

and $P (-1.96 < Z < 1.96) = 0.95$ [from normal probability tables]

$$\Rightarrow P \left(-1.96 < \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} < 1.96 \right) = 0.95$$

Thus, $\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}$ are the 95% confidence limits for the unknown parameter μ – the population mean and the interval

$\left[\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}} \right]$ is called the 95% confidence interval.

31.2.5 One -Tailed and Two-Tailed Tests

In any test, the critical region is represented by a portion of the area under the probability curve of the sampling distribution of the test statistic.

A test of any statistical hypothesis where the alternative hypothesis is one-tailed (right-tailed or left-tailed) is called a ‘one-tailed test’. For example, a test for testing the mean of a population $H_0: \mu = \mu_0$ against the alternative hypothesis: $H_1: \mu > \mu_0$ (right -tailed) or $H_1: \mu < \mu_0$ (left-tailed), is a ‘single-tailed test’. In the right-tailed test ($H_1: \mu > \mu_0$), the critical region lies entirely in the right tail of the sampling distribution of $\bar{x}$ (Fig 31.1), while for the left-test ($H_1: \mu < \mu_0$), the critical region is entirely in the left tail of the distribution (Fig 31.2).

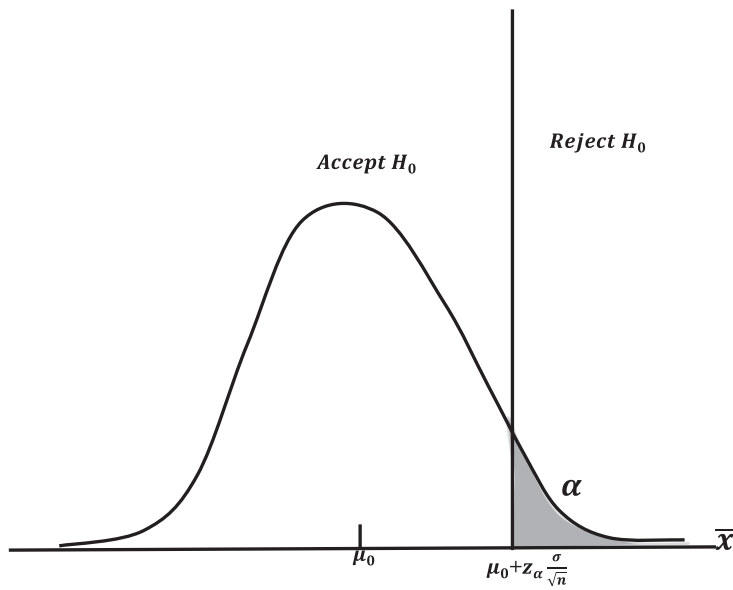


Fig. 31.1: Critical Region for One-Tailed Test ($H_1: \mu > \mu_0$)

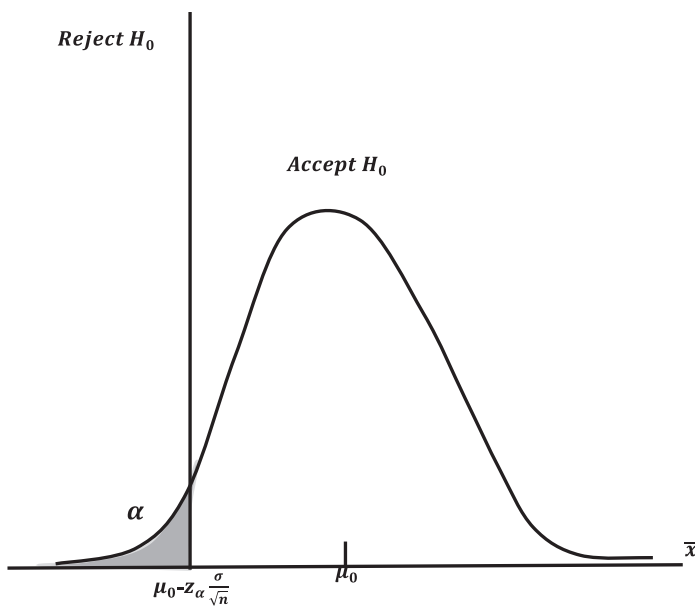


Fig. 31.2: Critical Region for One-Tailed Test ($H_1: \mu < \mu_0$)

Let us consider a test of statistical hypothesis where the alternative hypothesis is two-tailed (Fig 31.3) such as: $H_0: \mu = \mu_0$, against the alternative hypothesis $H_1: \mu \neq \mu_0$ (indicates both the possibilities $\mu > \mu_0$ and $\mu < \mu_0$). Such a test is known as ‘two-tailed test’ and in such a case the critical region is given by the portion of the area lying in both the tails of the probability curve of the test statistic.

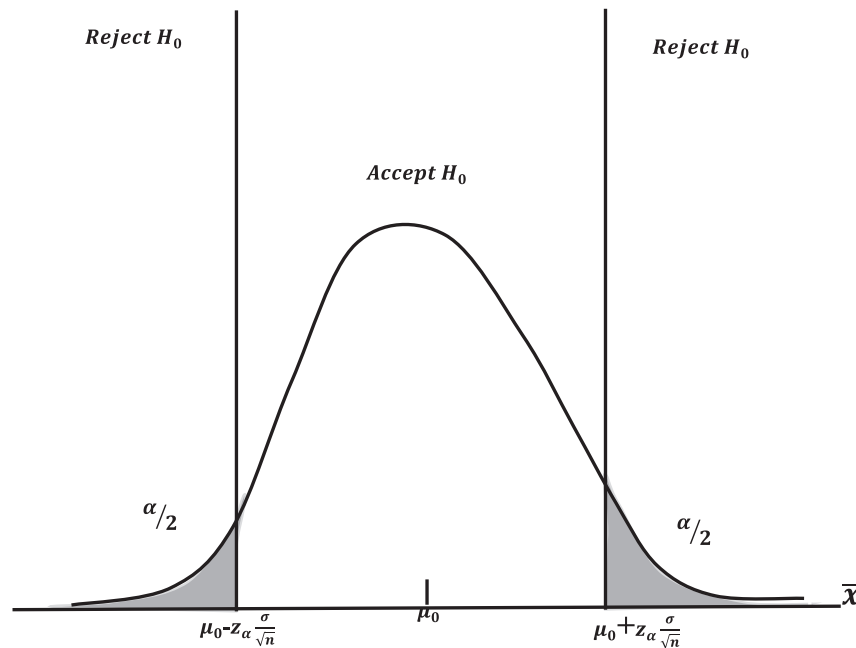


Fig. 31.3: Critical Region for Two-Tailed Test

In a particular problem, whether one-tailed or two-tailed test is to be applied depends entirely on the nature of the alternative hypothesis. If the alternative hypothesis is two-tailed, we apply the two-tailed test and if the alternative hypothesis is one-tailed, we apply the one-tailed test.

For example, suppose there are two popular brands of bulbs, one manufactured by standard process (with mean life μ_1) and the other manufactured by some new technique (with mean life μ_2). To test if the bulbs differ significantly, our null hypothesis is $H_0: \mu_1 = \mu_2$ and the alternative will be $H_1: \mu_1 \neq \mu_2$, thus giving us a two-tailed test. However, if we want to test if the bulbs produced by new process have higher average life than those produced by standard process, then we have $H_0: \mu_1 = \mu_2$ and $H_1: \mu_1 < \mu_2$, thus giving us a left-tailed test. Similarly, for testing if the product of new process is inferior to that of standard process, we have to set: $H_0: \mu_1 = \mu_2$ and $H_1: \mu_1 > \mu_2$, thus giving us a right-tailed test. Accordingly, the decision about applying a two-tailed test or a single-tail test (right or left) will depend on the problem under study.

31.2.6 Critical Values or Significant Values

The value of the test statistic which separates the critical (or rejection) region and the acceptance region is called the 'critical value' or 'significant value'. It depends upon: (i) the level of significance used, and (ii) the alternative hypothesis, whether it is two-tailed or single-tailed. As has been pointed out earlier, for large samples, the standardized variable asymptotically corresponding